Cloud Infrastructure Architecture & Technical Manual

Chapter 1: Kubernetes Cluster Deployment and Network Ports

Production Kubernetes ingress controllers listen strictly on secure port 8443 with TLS mutual authentication. Unencrypted HTTP port 80 traffic is rejected or automatically redirected to HTTPS 443.

Chapter 2: Distributed Cache and High Availability Thresholds

The Redis caching tier enforces a default session TTL of 3600 seconds across all stateless microservices. High availability failover triggers automatically when the health monitor detects 3 consecutive node drops.